

Message Code Macro name	Msq Code	bLength resultdata :	message transfer DATA[0] resunitdata DATA[1] DATA[0]	message transfer DATA[1] resunitdata DATA[2]	message transfer DATA[2] resunitdata DATA[3]	message transfer DATA[3] resunitdata DATA[4]	message transfer DATA[4] resunitdata DATA[5] (description)	message transfer DATA[5] resunitdata DATA[6] (description)	node/unit type	Remark	APDU service	Msq
EVENT MESSAGES												
1. UNIT OBJECT BINDING SPECIFIC EVENTS (Binding configuration with PowerMate PRO)												
EV_UnitControlToggle	01h	3	Binding nr MSB (binding NR = 0x0000 = special situation (for use with modem))	Binding nr LSB (binding NR = 0x0000 = special situation (for use with modem))	0 1		switch open switch closed after dt		Control unit	Status saved local response EV_BindingResponse response EV_UnitXXXStatus with changed attr. val.	if binfing NR != 0, explicit retry else implicit invoke	L/N
EV_UnitControlPulsToggle	02h	3	Binding nr MSB (binding NR = 0x0000 = special situation (for use with modem))	Binding nr LSB (binding NR = 0x0000 = special situation (for use with modem))	0 1		toggle puls Off toggle puls On		Control unit	Status saved local response EV_BindingResponse response EV_UnitXXXStatus with changed attr. val.		L/N
EV_UnitControlPuls	03h	2	Binding nr MSB (binding NR = 0x0000 = special situation (for use with modem))	Binding nr LSB (binding NR = 0x0000 = special situation (for use with modem))			pulsevent		Control unit	Saved as reference response EV_BindingResponse response EV_UnitXXXStatus with changed attr. val.		L/N
EV_UnitSelector	14h	3..5	Binding nr MSB (binding NR = 0x0000 = special situation (for use with modem))	Binding nr LSB (binding NR = 0x0000 = special situation (for use with modem))	Object member attribute NR Object member method NR	[val MSB] 00 toggle/pulstoggle 0 01 toggle/pulstoggle 1 10 puls Value Value Value	[val LSB] 00 toggle/pulstoggle 0 01 toggle/pulstoggle 1 10 puls		Selector unit	No reference used at destination. No logical constr. response EV_BindingResponse	if binfing NR != 0, explicit retry else implicit invoke	L/N
EV_BindingResponse	08h	3	Binding nr MSB	Binding nr LSB	BINDING_CHECK_OFF 0x00 BINDING_CHECK_ON 0x01 BINDING_CHECK_PULS 0x02 BINDING_CHECK_ERROR 0x03 BINDING_CHECK_NO_CHANGE 0x04 LAST IN BINDING SEQUENCE 0x10				unit		Result	L/N
2. UNIT OBJECT SPECIFIC EVENTS (Will be returned to all bReferencesToReturn adresses when an member attribute is changed)												
EV_UnitControlStatus	04h	3	5 = Control status	bUnitStatus1 bit 0 toggle status bit 1 pulstoggle status bit 2 pulstoggetimer running bit 3 reserved bit 4.5 state bit 6.7 ERROR code	bUnitStatus2 bit0 HWPorStatus (IN) bit 1 HWPorStatus (OUT) (ControlCheck) bit 2 controlchecked timer running bit 3 ControlPuls timer expired bit 4.6 reserved				Control unit			L/N
EV_UnitDimStatus	05h	3 3 2 2 2	1 = Dim speed 2 = Dim range 3 = Dimmer Value 4 = Dimmer ON/OFF 13 = Dimmer Current&Previous state	speed up (1..255 sec.) upper limit (1 .. 99) Value (0..99%) I/O bit 0.1 current status off.on.up.dn bit 2.3 previous status off.on.up.dn bit 4 PIR ON (in OFF state) bit 6 HW feedback error bit 0 HW feedback enable/disable bit 1,2 : 00=Triac PWM drive (factory reset) 01 = Triac continue drive (SSU)	speed dn (1..255 sec.) lower limit (1 .. 99)				Dim unit	Result	L/N	
		2	14 = Dimmer config (eeprom)									
		3	0x14 = DMX_Offset Address	Address MSB	Address LSB							
		4	0x15 = CFG Address	0x00 = Cfg GroupAddress	Address MSB	Address LSB						
EV_UnitSwitchStatus	06h	2 2	1 = Switch ON/OFF 5 = Switch current&previous state	I/O bit 0 current state off_on bit 1 previous state off_on bit 4 PIR ON (in OFF state) bit 6 HW feedback error bit 0 HW feedback enable/disable					Switch unit		Result	L/N
		2	6 = Switch config									
EV_UnitSensStatus	07h	4	1 = Working SetPoint	ValueMSB	ValueLSB	bit 6.5 Sensor preset 0 = SUN 1 = HALFSUN 2 = MOON 3 = HALFMOON bit 6.5 Sensor preset 0 = SUN 1 = HALFSUN 2 = MOON 3 = HALFMOON			Sens unit		Result	L/N
		4	2 = Hysteresis values	HystHighValue	HystLowValue							
		2 2 1	3 = Sensor control ON / OFF 7 = Sens sample interval 8 = Sens config	I/O Value bit 0.1, 0 = Temperatur (C) bit 0.1, 1 = Zuurtegraad (PH) bit 0.1, 2 = Lichtsterkte (Lumen) bit 0.1, 3 = Stroom (Ampère)								
		3 2	0xA = Current Measured Value 0xB = Sens Current&Previous state	ValueMSB bit 0.1 current status stable.high.low bit 2.3 previous status stable.high.low bit 4.5 Currentstatus	ValueLSB bit 0.1 current status stable.high.low bit 2.3 previous status stable.high.low bit 4.5 Currentstatus							
		3 2 3 3 2 2 3 2 3 3 3 3	0xC = Hysteresis offset values (always pos.) 0xD = PresetNR 0xE = Measured offset 0xF = NTC version info 0x10 = Workingsmode 0x11 = Fan Speed 0x12 = Swing angle 0x13 = Swing Mode 0x14 = Attr Set Delegate 0x15 = Attr Set Delegator 0x16 = Attr Set Request Measure value	HystHighOffset HystLowOffset bPresetNR (0..3) bMsb (short) bMsb (short) Mode Fan Speed bMsb (short) 0=off, 1=on bNodeAddress bNodeAddress bNodeAddress	HystLowOffset bLsb (short) bLsb (short) bLsb (short)				since v1.8 since v1.8 since v1.9 (Only Temperature Gateway) since v1.9 (Only Temperature Gateway) since v1.9 (Only Temperature Gateway) since v1.9 (Only Temperature Gateway) Only for Oled Room-e & Smartbox+ Only for Oled Room-e & Smartbox+ Only for Modbus sensor unit			
EV_NodeLCDStatus	19h								LCD node		Result	L/N
EV_UnitRC5IRReceiverStatus	22h	3 2	0x03 = State 0x02 = Config	bRC5RxStatus1 bRC5RxConfig 0 = RC5 1 = SONY	bRC5RxStatus2				RC5Rx unit (only in std nodes)		Result	N

TO DO

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EV_UnitAudioStatus (IR AUDIO all configurations or Bose audio)	17h	2 2 2	0x01 = ATTR_SOURCE_ON 0x06 = ATTR_ONOFF 0x10 = ATTR_STATE	0.5 = source, 0xFF = NOT RELEVANT 0 = OFF, 1 = ON bAudioStatus1 (bit 0..3) AUDIO_UNIT_OFF_STATE 0x00 AUDIO_UNIT_ON_STATE 0x01 AUDIO_UNIT_VOLUP_STATE 0x02 AUDIO_UNIT_VOLDOWN_STATE 0x03 PREVIOUS_AUDIO_OFF 0x00 PREVIOUS_AUDIO_ON 0x04 PREVIOUS_AUDIO_VOLUP 0x08 PREVIOUS_AUDIO_VOLDOWN 0x0C						Audio unit in LCD node	Result	L/N
		2 2 2	0x0D = ATTR_CONFIG 0x10 = ATTR_STATE 0x11 = ATTR_DISC (Bose)	bAudioConfig bAudioStatus1 bCDTrackDisc bit 7.5 Disc NR bit 4.0 Track NR LSB per 1 Mc								
		4 2	0x12 = ATTR_TUNERFREQ (Bose) 0x13 = ATTR_VOLUME (Bose)	bCurrentVolume	MSB byte 50 Kc	bPresetNR (0..n) 0xFF=manual				Only with Bosers unit Only with Bosers unit		
EV_UnitAudioStatus (MATRIX AUDIO HC4 and HC5)	17h	3 3 3 2 2 6	0x01 ATTR_SOURCE_ON 0x06 ATTR_ONOFF 0x0C ATTR_VOLUME 0x0D ATTR_CONFIG 0x0E ATTR_STATE 0x16 ATTR_DEST_STATUS 22 / R data 5 bytes	bAudioDestNR bAudioDestNR bAudioDestNR bAudioConfig bAudioStatus1 bCurrentDestination	bAudioSourceNR bState 0/1 bDestinationVolume							
		5	0x17 ATTR_VOLUME_STATUS	bDestinationVolume0	bit 7 assigned/not assigned bit 6 on/off bit 5..0 source 0..63 bDestinationVolume1	bDestinationState1 bit 7 assigned/not assigned bit 6 on/off bit 5..0 source 0..63 bDestinationVolume2	bDestinationState2 bit 7 assigned/not assigned bit 6 on/off bit 5..0 source 0..63 bDestinationVolume3	bDestinationState3 bit 7 assigned/not assigned bit 6 on/off bit 5..0 source 0..63				
EV_UnitDuoSwitchStatus	26h	2 2 2	0x01 = UP / STOP 0x02 = DOWN / STOP 0x07 = State	1=UP, 0=STOP 1=DOWN, 0=STOP bDuoSwitchStatus1 bit 0..2 current state bit 3..5 previous state bit 6..7 error code DUOSWITCH_UNIT_STOP_STATE 0x00 DUOSWITCH_UNIT_WAIT_UP_STATE 0x01 DUOSWITCH_UNIT_WAIT_DOWN_STATE 0x02 DUOSWITCH_UNIT_UP_STATE 0x03 DUOSWITCH_UNIT_DOWN_STATE 0x04 DUOSWITCH_UNIT_STOP_PREVIOUS_STATE 0x00 DUOSWITCH_UNIT_WAIT_UP_PREVIOUS_STATE 0x08 DUOSWITCH_UNIT_WAIT_DOWN_PREVIOUS_STATE 0x10 DUOSWITCH_UNIT_UP_PREVIOUS_STATE 0x18 DUOSWITCH_UNIT_DOWN_PREVIOUS_STATE 0x20								
		2 2	0x08 = config 0x09 = stop timer reload value	bConfig bStopTimerReloadValue (1 = 0.6 sec.) factory setting = 100 (60 sec.)								
		2	0x0A = switch timer reload value	bSwitchTimerReloadValue (1 = 0.06 sec.) factory setting = 10 (600 msec.)								
EV_UNITIRTSTATUS	2Dh	2 2	0x04 = CONFIG 0x05 = STATE	bIRTXConfig bIRTxStatus1 STATE_IRTX_IDLE 0x00 STATE_IRTX_REQUEST_IRCODE 0x01 STATE_IRTX_CHAIN_IRTX 0x02 STATE_IRTX_WAIT_FOR_NEXT_IRCODE 0x03 PREVIOUS_STATE_IRTX_IDLE 0x00 PREVIOUS_STATE_IRTX_REQUEST_IRCODE 0x04 PREVIOUS_STATE_IRTX_CHAIN_IRTX 0x08 PREVIOUS_STATE_IRTX_WAIT_FOR_NEXT_IRCODE 0x0C								
EV_UnitSelectorStatus	30h	2 2	0x01 = SELECTOR_OBJECT_ATTR_STATE 0x02 = SELECTOR_OBJECT_ATTR_TYPE	bUnitSelectorStatus1 bUnitType (see EV_UnitConfig)								
EV_UnitVideoMuxStatus	31h	4 2	0x01=OBJECT_ATTR_CONNECTION 0x06=OBJECT_ATTR_CONFIG	bOutput (0..MAX-1) bVideoMuxConfig	bInput (0..MAX-1)	bState (1=connect,0=disconnect)					result	L/N
EV_UnitPS2IntelliMouseStatus	32h	5 2	0x08=OBJECT_ATTR_SCREENSIZE 0x0A=OBJECT_ATTR_CONFIG	bXMsb bPS2IntelliMouseConfig	bXLsb	bYMsb	bYLsb				result	L/N
EV_UnitAVMatrixSet	36h	1 4 3 3 3 2 2 2 2 5	0x00 OBJECT_USE_SELECTOR_DATA 0x01 OBJECT_ATTR_CONNECTION 0x04 OBJECT_ATTR_VOLUME 0x05 OBJECT_ATTR_BASS 0x06 OBJECT_ATTR_TREBLE 0x0D AVMATRIX_OBJECT_ATTR_STATE 0x0E AVMATRIX_OBJECT_ATTR_CONFIG 0x0F OBJECT_CONFIG_NR_OF_DEST_ON_UNITS 0x10 OBJECT_ATTR_CONFIG_NO_OF_UNITS 0x13 OBJECT_METHOD_MACRO_RECEIVED	bMatrixOutput bMatrixOutput bMatrixOutput bMatrixOutput bAVMatrixStatus1 bAVMatrixConfig nr. of destinations on 1 unit (0x00 - 0x3F) nr. of units (0x00 - 0x0F) Dest/Room NR	bMatrixInput bVolume bBass bTreble	bState (1/0)			Response to FC UnitAVMatrixSet	Result	L/N	
					function MSB	function Lsb	0 Event toggle off 1 Event toggle on 2 Event Puls					
EV_UnitAlarmStatus	37h	2 5 2 2 2 2 3 1 1 2 2	0x03=OBJECT_ATTR_INFO 0x04=OBJECT_ATTR_GROUPSTATUS 0x05=OBJECT_ATTR_ALARMZONE 0x06=OBJECT_ATTR_OPENZONE 0x07=OBJECT_ATTR_PRECOUNTER 0x08=OBJECT_ATTR_REMOTECODE 0x09=OBJECT_ATTR_CONFIG 0x0B=OBJECT_ATTR_STATE 0x0C=OBJECT_ATTR_REMOTECODE_READ_ERROR 0x0D=OBJECT_ATTR_REMOTECODE_WRITE_OK 0x0E=OBJECT_ATTR_REMOTECODE_WRITE_ERROR 0x0F=OBJECT_ATTR_OPENAREA_EXITTIME 0x10=OBJECT_ATTR_OPENAREA_ENTRYTIME	0x00..0x0F bGroup1 0..207 0..207 0..30 bIndex (0..7) 0=Galaxy60 1=Galaxy500 bAlarmStatus1 bAlarmStatus2	bGroup2	bGroup3	bGroup4	Char	Char	Char	Only Alarm Aritech (since v1.7) Only Alarm Aritech (since v1.7)	

[illegible]

TO DO

Message Code	Msq	length	message transfer DATA[0]	message transfer DATA[1]	message transfer DATA[2]	message transfer DATA[3]	message transfer DATA[4]	message transfer DATA[5]	message transfer DATA[6]	node unit	Remark	APDU service	Msq
Macro name	Code	result data : DATA[0]	result data DATA[1]	result data DATA[2]	result data DATA[3]	result data DATA[4]	result data DATA[5] (description)	result data DATA[6] (description)	type				
	1		0x33=MOOD CONFIG INITIALIZE ERROR										
	1		0x35=MOODHEADER CONFIG_SET_OK										
	1		0x36=MOODHEADER CONFIG_SET_ERROR										
	1		0x38=MOODENTRY CONFIG_SET_OK										
	1		0x39=MOODENTRY_CONFIG_SET_ERROR										
	1		0x3B=MOOD CONFIG DONE_OK										
	1		0x3C=MOOD_CONFIG_DONE_ERROR										
	1		0x3E=IRRX CONFIG INITIALIZE_OK										
	1		0x3F=IRRX_CONFIG_INITIALIZE_ERROR										
	1		0x41=IRRX CONFIG_SET_OK										
	1		0x42=IRRX_CONFIG_SET_ERROR										
	1		0x44=IRRX CONFIG_DONE_OK										
	1		0x45=IRRX_CONFIG_DONE_ERROR										
	1		0x47=SCHEDULE_CONFIG_INITIALIZE_OK										
	1		0x48=SCHEDULE CONFIG_INITIALIZE_ERROR										
	1		0x4A=SCHEDULEITEM_CONFIG_SET_OK										
	1		0x4B=SCHEDULEITEM CONFIG_SET_ERROR										
	1		0x4D=SCHEDULEEVENT_CONFIG_SET_OK										
	1		0x4E=SCHEDULEEVENT CONFIG_SET_ERROR										
	1		0x50=SCHEDULE_CONFIG_DONE_OK										
	1		0x51=SCHEDULE CONFIG_DONE_ERROR										
	1		0x53=SMAP_CONFIG_INITIALIZE_OK										
	1		0x54=SMAP CONFIG_INITIALIZE_ERROR										
	1		0x56=SMAP CONFIG_PAGEINFO_HEADER_OK										
	1		0x57=SMAP_CONFIG_PAGEINFO_HEADER_ERROR										
	1		0x59=SMAP CONFIG_PAGEINFO_DATA_OK										
	1		0x5A=SMAP_CONFIG_PAGEINFO_DATA_ERROR										
	1		0x5C=SMAP CONFIG_DONE_OK										
	1		0x5D=SMAP_CONFIG_DONE_ERROR										
	1		0x5F=SMAP MENU CONFIG_INITIALIZE_OK										
	1		0x60=SMAP_MENU_CONFIG_INITIALIZE_ERROR										
	1		0x62=SMAP_MENU_CONFIG_SET_OK										
	1		0x63=SMAP_MENU_CONFIG_SET_ERROR										
	1		0x65=SMAP MENU CONFIG_DONE_OK										
	1		0x66=SMAP MENU CONFIG_DONE_ERROR										
	1		0x71=Gateway_Cfg_Univ_Initialize_Ok										
	1		0x72=Gateway_Cfg_Univ_Initialize_Error										
	1		0x74=Gateway_Cfg_Univ_Set_Ok										
	1		0x75=Gateway_Cfg_Univ_Set_Error										
	1		0x77=Gateway_Cfg_Done_Ok										
	1		0x78=Gateway_Cfg_Done_Error										
	1		0x81=RoomCtrl_AudioCfg_Initialise_Ok										
	1		0x82=RoomCtrl_AudioCfg_Initialise_Error										
	1		0x82=RoomCtrl_AudioCfg_Dest_Set_Ok										
	1		0x83=RoomCtrl_AudioCfg_Dest_Set_Error										
	1		0x87=RoomCtrl_AudioCfg_Src_Set_Ok										
	1		0x88=RoomCtrl_AudioCfg_Src_Set_Error										
	1		0x8A=RoomCtrl_AudioCfg_TmrPreset_Set_Ok										
	1		0x8B=RoomCtrl_AudioCfg_TmrPreset_Set_Error										
	1		0x8D=RoomCtrl_AudioCfg_SrcFunc_Set_Ok										
	1		0x8E=RoomCtrl_AudioCfg_SrcFunc_Set_Error										
	1		0x90=RoomCtrl_AudioCfg_Done_Ok										
	1		0x91=RoomCtrl_AudioCfg_Done_Error										
	1		0xA1=AudioRender_Cfg_Initialise_ok										
	1		0xA2=AudioRender_Cfg_Initialise_Error										
	1		0xA5=AudioRender_Cfg_Render_Ok										
	1		0xA5=AudioRender_Cfg_Render_Error										
	1		0xA7=AudioRender_Cfg_Done_ok										
	1		0xA8=AudioRender_Cfg_Done_Error										
	1		0xB1=LedCfg_Init_Ok										
	1		0xB2=LedCfg_Init_Error										
	1		0xB4=LedCfg_Data_Ok										
	1		0xB5=LedCfg_Data_Error										
	1		0xB7=LedCfg_Data_Led_Ok										
	1		0xB8=LedCfg_Data_Led_Error										
	1		0xBA=LedCfg_Done_Ok										
	1		0xBB=LedCfg_Done_Error										
	1		0xC1=SmapiLite_init_ok										
	1		0xC2=SmapiLite_init_error										
	1		0xC4=SmapiLite_entry_ok										
	1		0xC5=SmapiLite_entry_error										
	1		0xC7=SmapiLite_done_ok										
	1		0xC8=SmapiLite_done_error										
	1		0xFF=EXTERNALCONFIG NOT IMPLEMENTED										
EV_FTPFILECONTROLDATA	24h	segmented	segmented data							node	response to FTP_FILECONTROL - Read data	Result	
EV_FTPFILECONTROLSTATUS	25h	5	0x00=OBJECT_OPEN_WRITE	HMSB Filelength (32bit)	HLSB Filelength (32bit)	LMSB Filelength (32bit)	LLSB Filelength (32bit)			node	response to FTP_FILECONTROL	Result	
		5	0x01=OBJECT_OPEN_READ	HMSB Filelength (32bit)	HLSB Filelength (32bit)	LMSB Filelength (32bit)	LLSB Filelength (32bit)						
		1	0x02=OBJECT_APPEND_DATA										
		5	0x03=OBJECT_CLOSE	HMSB Filelength (32bit)	HLSB Filelength (32bit)	LMSB Filelength (32bit)	LLSB Filelength (32bit)						
		5	0x04=OBJECT_INFO_FILESIZE	HMSB Filelength (32bit)	HLSB Filelength (32bit)	LMSB Filelength (32bit)	LLSB Filelength (32bit)						
		1	0x05=OBJECT_ABORT										
		1	0x80=OBJECT_EC_WRONG_STATE										
		1	0x81=OBJECT_EC_WRITE_DATA										
		1	0x82=OBJECT_EC_READ_DATA										
		1	0x83=OBJECT_EC_OPEN_FILE										
		1	0x84=OBJECT_EC_CLOSE_FILE										
		1	0x85=OBJECT_EC_GENERAL										
		1	0x86=OBJECT_EC_NOT_IMPLEMENTED										
		1	0x87=OBJECT_EC_GET_INFO										
EV_SYSTEMBRIDGEMESSAGE	3Ch		0x00=ATTR_DALI_SUBSCRIBE (RESERVED)										
		2	0x02=OBJECT_DALI_RESPONSE_DATA	Backward response data							Dali/CAN bridge	Implicit	
		2	0x03=OBJECT_DALI_RESPONSE_CODE	Response code									
			0x04=OBJECT_DALI_EVENT (RESERVED)										

6. NODE OBJECT BINDING CONFIGURATION EVENTS

Message Code Macro name	Msq Code	bLength resultdata : DATA[0]	message transfer DATA[0] resultdata DATA[1]	message transfer DATA[1] resultdata DATA[2]	message transfer DATA[2] resultdata DATA[3]	message transfer DATA[3] resultdata DATA[4]	message transfer DATA[4] resultdata DATA[5] (description)	message transfer DATA[5] resultdata DATA[6] (description)	node/unit type	Remark	APDU service	Msq
EV_NodeBindingEntry	10h	4 1 PackLn 3 1 2 1 PackLn 1 1 1 1 1 1 1 1 1 1 1 1	0 = Header read 1 = Header written 2 = Data read 3 = Data written 4 = Entry cleared 5 = Write done 6 = Write/Read cancelled 7 = Read done 80 = Wrong binding entry NR 81 = Write error wrong state 82 = Read error wrong state 83 = Binding file full 84 = Entry error 85 = Binding file closed 86 = Binding file empty 87 = Binding file corrupt 88 = Binding config wrong state 89 = Binding temp. buffer full	Binding NR MSB Data Nr of Binding entries Binding Entry NR Data	Binding NR LSB Data XOR file (B,N,C .. S) Data	Feedback/Response (see FC_xxx) Data Data	 Data Data	 Data Data	node	Binding configuration (PinInstall)	Result	N
EV_NodeBindingsFileControl	13h	3 3 3 1 1 1 1	0 = File Closed 1 = File Opened 2 = File Info 3 = File erased 80 = Close Error 81 = Open Error 82 = Share Violation 83 = Binding temp buffer in use	Nr of Binding entries Nr of Binding entries Nr of Binding entries	XOR file (B,N,C .. S) XOR file (B,N,C .. S) XOR file (B,N,C .. S)				node	Binding configuration (PinInstall)	Result	N
7. NODE PROTOCOL SPECIFIC EVENTS												
EV_MessageError	11h	2	bErrorNR = MESSAGE_ERROR_REFUSED_RUNNING_STATE 0x00 MESSAGE_ERROR_INVALID 0x01 MESSAGE_ERROR_BUFFER_FULL 0x02 MESSAGE_ERROR_BINDING_NOT_FOUND 0x03 MESSAGE_ERROR_REFERENCE_NOT_FOUND 0x04 MESSAGE_ERROR_BINDING_TYPE_INVALID 0x05 MESSAGE_ERROR_MESSAGE_TRANSFER_TIMEOUT 0x06 MESSAGE_ERROR_DLL_ACK_TIMEOUT 0x07 MESSAGE_ERROR_TRANSCIEVER_CHIP 0x08 MESSAGE_ERROR_UNIT_NR_INVALID 0x09 MESSAGE_ERROR_BINDINGCONFIG_REFUSED 0x0A MESSAGE_ERROR_WRONG_OBJECT_MEMBER 0x0B MESSAGE_ERROR_INVALID_MESSAGE_CODE 0x0C MESSAGE_ERROR_WRONG_DIRECT_CALL 0x0D MESSAGE_ERROR_REFUSED_WRONG_DATA 0x0E MESSAGE_ERROR_REFUSED_BINDING_STATE 0x0F MESSAGE_ERROR_CAN_RX_PROTOCOL_TIMEOUT 0x10 MESSAGE_ERROR_CAN_TX_PROTOCOL_TIMEOUT 0x11 MESSAGE_ERROR_ONLY_FOR_MASTER_LCD 0x12 MESSAGE_ERROR_NO_FREE_MTC_MAX_RETRIES 0x13 MESSAGE_ERROR_BINDING_SYNTAX_ERROR 0x14 MESSAGE_ERROR_BINDING_MAX_CONDITIONS 0x15 MESSAGE_ERROR_BINDING_MAX_TIMEROPERATORS 0x16 MESSAGE_ERROR_BINDING_MAX_ENTRIES 0x17 MESSAGE_ERROR_BINDING_CONDITION_LENGTH 0x18 MESSAGE_ERROR_BINDING_MAX_REFERENCES 0x19 MESSAGE_ERROR_BINDING_BRACES_SYNTAX 0x1A	bNodeCurrentState						unit	Implicit Invoke / result	L/N
EV_NodeDLLStatus (Powerline)	18h	6 4	Nic error block NR 0 1	lower nibble --> upper nibble --> lower nibble --> upper nibble -->	2 nibbles / byte data NIC_ERROR_RESET NIC_ERROR_LINK_STATUS NIC_ERROR_INT_STATE NIC_ERROR_LAYER_CONFIG	2 nibbles / byte data NIC_ERROR_LINK_STATUS_RESET NIC_ERROR_LINK_STATUS_INFO_RESET NIC_ERROR_WRONG_PACKET_LENGTH NIC_ERROR_READSEQUENCE	2 nibbles / byte data NIC_ERROR_INTERFACE NIC_ERROR PHYSICAL NIC_ERROR RC EXCEPT RESET	2 nibbles / byte data NIC_ERROR_RC_EXCEPT NIC_ERROR TX EXCEPT RESET		Implicit Invoke	N	
EV_NodeDLLStatus (CAN)	18h	6 4	Nic error block NR 0 1	lower nibble --> upper nibble --> lower nibble --> upper nibble -->	2 nibbles / byte data NIC_ERROR_RESET NIC_FLOW_TIMEOUT NIC_ERROR_HOST_BUSY NIC_ERROR_NO_FREE_MTC	2 nibbles / byte data NIC_TX WARNING NIC_RX PASSIVE NIC_ERROR FRAME ERROR NIC_ERROR_SEGMENT_ERROR	2 nibbles / byte data NIC TX PASSIVE NIC_ERROR_TX_BUSOFF NIC_ERROR SEGMSG TIMEOUT TEC	2 nibbles / byte data NIC_ERROR RX0 OVERRUN NIC_ERROR_RX1_OVERRUN NIC_ERROR_DLL_TIMEOUT		Implicit Invoke	N	
EV_Init	12h	1	0x02,0x03 = Power ON reset 0x00 = Watchdog reset 0x01 = External reset 0x04 = FC_NODERESET 0x05 = Unit reset							>20s. Puls on P&P button	Implicit Invoke	N
8. NODE OBJECT EEPROM SPECIFIC MANAGEMENT EVENTS												
EV_NodeEEPROMWritten	7Fh	0										
EV_NodeEEPROMRead	7Eh	1	Val									
EV_NodeEEPROMErased	15h	0										
EV_DatabaseManagement	1Eh	1 1 1 1 1	0x00 OBJECT_ERASE 0x01 OBJECT_LOAD 0x02 OBJECT_SAVE 0x03 OBJECT_BACKUP 0x80 EC_NOT_IMPLEMENTED						node	response to FC_DatabaseManagement. since v65.24	Only in NODESTATUS_RUNNING stat result N	N
9. LCD NODE OBJECT SPECIFIC EVENTS												
EV_SlaveLCDBrowsingDone	1Ch	1	SUCCESS = 1 ERROR = 0								Only in NODESTATUS_RUNNING stat	N
EV_NodeLCDMasterStatus	1Dh	2 2	0x00 = Weekschedule 0x02 = Alarm Code	bWeekScheduleNR (0..3) (0=Wrong code, 1=Code OK)							Only in NODESTATUS_RUNNING stat	N

TO DO

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		2	0x03 = Alarm Status	ALARM_ZONE_A = 0x01 ALARM_ZONE_B = 0x02 ALARM_ACTION = 0x10	Node address(15..8)	Node address(7..0)	bUnitAddress					
		5	0x04 = Alarm Event	ALARM_ZONE_A = 0x01 ALARM_ZONE_B = 0x02 ALARM_ACTION = 0x10	Node address(15..8)	Node address(7..0)	bUnitAddress					
		6	0x07 = Registermap address 0x0000 - 0x00ff	RegisterAddress	HMSB value	HLSB value	LMSB value	LLSB value				
		6	0x08 = Registermap address 0x0100 - 0x01ff	RegisterAddress	HMSB value	HLSB value	LMSB value	LLSB value				
		6	0x09 = Registermap address 0x0200 - 0x02ff	RegisterAddress	HMSB value	HLSB value	LMSB value	LLSB value				
		6	0x07 = Registermap address 0x0300 - 0x03ff	RegisterAddress	HMSB value	HLSB value	LMSB value	LLSB value				
EV_MemoryDataDump		7Ch	Segmented data						LCD node	response to FC_GetMemoryData	result	L/N
EV_WriteMemoryDataDone		7Dh	0						LCD node	response to FC_WriteMemoryData	result	L/N
EV_LCDPMDbaseAddNodeEntry		29h	0						LCD node	response to FC_LCDPMDbaseAddNodeEntry	result	L/N
EV_LCDPMDbaseAddUnitEntry		2Ah	0						LCD node	response to FC_LCDPMDbaseAddUnitEntry	result	L/N
EV_IRCODEDATA		2Ch	Segmented data IR code following structure (BIG ENDIAN)						LCD node	response to FC_GETIRCODE from IRTX unit	APDU_RESULT	N
EV_IRCODEDATABASEERROR		2Fh	0						LCD node	response to FC_GETIRCODE from IRTX unit if a code wasn't found or an error happened !	APDU_RESULT	N
EV_LCDINFODIALOG		2Bh	0						LCD node	response to FC_LCDINFODIALOG	APDU_RESULT	N
EV_RC5CODEMATCH		2Eh	2	bMapIndex 0..63 (used in bit 0.5 in binding flags for B and I binding)	bEventCode 0 = RC5_TOGGLE_OFF_EVENT 1 = RC5_TOGGLE_ON_EVENT 2 = RC5_PULS_EVENT 3 = RC5_NO_EVENT				LCD node	response to FC_CHECKRC5CODE from RC5Rx unit	APDU_RESULT	N
EV_LCDBUILDSCREENMAP		34h	1	SCREENMAP_INIT_OK								
			1	SCREENMAP_INIT_ERROR								
			1	SCREENMAP_ENTRY_OK								
			1	SCREENMAP_ENTRY_ERROR								
			1	SCREENMAP_NODE_NOT_FOUND								
			1	SCREENMAP_UNIT_NOT_FOUND								
EV_NodeGatewayStatus		7Bh	2	0x01 = ATTR_STATE	0x00 = DATABASE_UPDATE_ERROR 0x01 = DATABASE_UPDATE_DONE 0x02 = DATABASE_UPDATE_STARTED				Gateway	response to FC_NodeGatewayRequestStatus	APDU_RESULT	
10. Reserved Events modem v65.24												
EV_NODEPNPREPONSE		0fh								Can be reused		
EV_UNITAUDIOCONTROL		16h								Can be reused		
EV_NODEIRCODEDATABASECHANGED		1fh								Can be reused		
EV_NODEAVIRDEVICESDATABASECHANGED		20h								Can be reused		
EV_NODEAVIRDEVICESENDIRCODE		21h								Can be reused		
EV_UNITRC5RRCCEIVERSTATUS		22h								Can be reused		
EV_NODEAVIRMATRIXSTATUS		23h								Can be reused		
EV_UNITSPEECHCOMMSTATUS		33h								Can be reused		
EV_Temp1		38h								Reserved		
EV_Temp2		39h								Reserved		
EV_Temp3		3Ah								Reserved		
EV_Temp4		3Bh								Reserved		
FUNCTION MESSAGES												
1. UNIT OBJECT FUNCTIONS TO BIND OR DIRECT CALL												
FC_UnitRequestDimStatus		83h	1	0x01 = Dim speed					Dim unit	Response : EV_UnitDimStatus	Explicit Invoke	L/N
			1	0x02 = Dim range								
			1	0x03 = Dim Value								
			1	0x0d = Dimmer Current&Previous state								
			1	0x0e = Dimmer config (eeprom)								
			1	0x14 = DMX_Offset Address								
			2	0x15 = CFG Address	0x00 = Cfg GroupAddress							
FC_UnitDimSet		A2h	1	0x00 = Use EV_UnitSelector data to set values					Dim unit	Response : EV_UnitDirectSet		
			3	0x01 = Dim speed	speed up (1..255 sec.)	speed up (1..255 sec.)						
			3	0x02 = Dim range	upper limit (1..99)	upper limit (1..99)						
			2	0x03 = Dim Value	Value (0..1..99,100)	lower limit (1..99)						
			2	0x04 = Dimmer ON/OFF	bit 0 : 1/0							
			1	0x05 = up								
			1	0x06 = down								
			1	0x07 = f(status), up or down								
			1	0x08 = f(status), up or down and switch on or off								
			1	0x09 = off								
			1	0x0a = Switch on with current dimmer value.								
			1	0x0b = f(status), current value on OR off								
			2	0x0c = f(status) dimmer value or off	Value (0..1..99,100)							
			2	0x0e = Dimmer config (eeprom)	bit 0 HW feedback enable / disable(factory reset) bit 1..2 : 00=Triac PWM drive (factory reset) (Not for DC dimmer) 01 = Triac continue drive (SSL) (Not for DC dimmer) bit 6 Pir-On-Function Delay Enable bit 7 Pir-Off-Function Delay Enable							
			1	0x0f = Increment (+5 step)								
			1	0x10 = Decrement (-5 step)								
			4	0x11 = Switch ON and dim UP	Value (0..1..99,100) if 0xFF, do not use speed up (1..255 sec.) if 0xFF, do not use 00 = stay in UP state							
			4	0x12 = Switch ON and dim DOWN	Value (0..1..99,100) if 0xFF, do not use speed dn (1..255 sec.) if 0xFF, do not use 00 = stay in down state							
			3	0x13 = PIR ON	Time (1..60 min OR 0xFF (= def. 3 min) Value (1..99% or 0xFF (=curr. Val.))							
			3	0x14 = DMX_Offset Address	Address MSB	Address LSB						
			4	0x15 = CFG Address	0x00 = Cfg GroupAddress	Address MSB						

TO DO

Message Code Macro name	Msg Code	length resultdata : DATA[0]	message transfer DATA[0] resultdata DATA[1]	message transfer DATA[1] resultdata DATA[2]	message transfer DATA[2] resultdata DATA[3]	message transfer DATA[3] resultdata DATA[4]	message transfer DATA[4] resultdata DATA[5] (description)	message transfer DATA[5] resultdata DATA[6] (description)	node/unit type	Remark	APDU service	Msg
FC_UnitRequestSwitchStatus	86h	1 1 1	0x01 = Switch ON/OFF 0x05 = Switch current&previous state 0x06 = Switch config (eeprom)						Switch unit	Response : EV_UnitDirectSet	Explicit Invoke	L/N
FC_UnitSwitchSet	A3h	1 2 1 1 1 2 2 2	0x00 = Use EV_UnitSelector data to set values 0x01 = Switch ON/OFF 0x02 = off 0x03 = on 0x04 = f(status), on or off 0x06 = Switch config (eeprom) 0x07 = PIR ON 0x08 = TIMER VALUE	bit 0 : 1/0					Switch unit	Response : EV_UnitDirectSet Response if status change : EV_UnitSwitchStatus	Explicit Invoke	
FC_UnitSensSet	88h	1 4 4 2 1 1 1 2 2 1 3 2 3 2 3 3 3	0x00 = Use EV_UnitSelector data to set values 0x01 = Working SetPoint (2 Bytes R/W) 0x02 = Hysteresis values (2 Bytes R/W) 0x03 = Sensor control on/off 0x04 = Use EV_Controlxxx to control SetPoint, 1:Inc setpoint, 0:Dec Setpoint (EV_Controlxxx data) 0x05 = Inc Setpoint 0x06 = Dec Setpoint 0x07 = Sens sample interval (1 Byte R/W) 0x08 = Sens Type (1 Byte R/W) (eeprom) 0x09 = f(status) Sensor control ON/OFF 0x0c = Hysteresisch offset 0x0d = Preset NR 0x0E = Measured offset value 0x10 = Workingsmode 0x11 = Fanspeed 0x12 = Swing angle 0x13 = Swing mode 0x14 = Attr Set Delegate 0x15 = Attr Set Delegator 0x16 = Attr Set Request Measure value	ValueMSB HystHighOffset 1/0 Value 0 = Temperatur (C) (factory reset) 1 = Zuurtegraad (PH) 2 = Lichtsterkte (Lumen) 3 = Stroom (Ampère) High 0 = SUN 1 = HALFSUN 2 = MOON 3 = HALFMOON bLsb (Short) bLsb (Short)	ValueLSB HystLowOffset bLsb (Short)	bPresetNR 0 = SUN 1 = HALFSUN 2 = MOON 3 = HALFMOON bPresetNR 0 = SUN 1 = HALFSUN 2 = MOON 3 = HALFMOON		Sens unit	Response : EV_UnitDirectSet	Explicit Invoke	L/N	
FC_UnitRequestSensStatus	89h	1/2 1/2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1	0x01 = Working SetPoint 0x02 = Hysteresis values 0x03 = Sensor control ON / OFF 0x07 = Sens sample interval 0x08 = Sens Type (eeprom) 0x0a = Current Measured Value 0x0b = Sens current&previous state 0x0c = Hysteresisch offset 0x0d = Preset NR 0x0E = Measured offset value 0x0F = NTC Version 0x10 = Workingsmode 0x11 = Fan Speed 0x12 = Swing angle 0x13 = Swing Mode 0x14 = Attr Set Delegate 0x15 = Attr Set Delegator 0x16 = Attr Set Request Measure value	(if length==2) bPresetNR (0..3) 0 = SUN 1 = HALFSUN 2 = MOON 3 = HALFMOON (if length==2) bPresetNR (0..3) 0 = SUN 1 = HALFSUN 2 = MOON 3 = HALFMOON					Sens unit	Response : EV_UnitSensStatus	Explicit Invoke	
FC_UnitControlSet	A8h	2 1 2/1 (if bind) 2/1 (if bind)	0x01 = Check 0x02 = Control PULS event 0x03 = Control PULSTOGGLE event 0x04 = Control TOGGLE event	0 : Check off 1 : Check on 2 : Check error 3 : Check puls bit 0 : 1/0 (when used as result of a C or G binding, length =1 and NO data) bit 0 : 1/0 (when used as result of a C or G binding, length =1 and NO data)								
FC_UnitRequestControlStatus	87h	1	0x05 = Control status						Control unit	Response : EV_UnitControlStatus	Explicit Invoke	L/N
FC_NodeLCDSet	A1h	1 2 2 3	0x00 = Use EV_UnitSelector data to set values 0x01 = TFT_VIDEO_CONTROL 0x02 = TFT_VIDEODECODER INPUT 0x03 = TFT_MOUSE_CONTROL	bMode EV_LCDTFTTOFF 0 EV_LCDTFTPIP 1 EV_LCDTFTFS 2 LCDTFT_DECODER CV 0 LCDTFT_DECODER SVHS 1 LCDTFT_DECODER RGB 2 bNodeAddressPS2IntelliMouse	bUnitAddressPS2IntelliMouse			LCD unit	Response : EV_NodeDirectSet	Explicit Invoke	L/N	
FC_UnitLCDVirtualSet	A6h	2 1	0x01 = Check 0x02 = Control PULS event	0 : Check off 1 : Check on 2 : Check error 3 : Check puls					LCD virtual unit	Response : EV_UnitDirectSet	Explicit Invoke	

TO DO

Message Code Macro name	Msg Code	length resultdata : DATA[0]	message transfer DATA[0] resultdata DATA[1]	message transfer DATA[1] resultdata DATA[2]	message transfer DATA[2] resultdata DATA[3]	message transfer DATA[3] resultdata DATA[4]	message transfer DATA[4] resultdata DATA[5] (description)	message transfer DATA[5] resultdata DATA[6] (description)	node/unit type	Remark	APDU service	Msg
		2	0x03 = Control PULSTOGGLE event	bit 0 : 1/0								
		2	0x04 = Control TOGGLE event	bit 0 : 1/0								
FC_UnitRequestLCDVirtualStatus	A7h	1	0x05 = Control status						LCD virtual unit	Response : EV_UnitLCDVirtualStatus	Explicit Invoke	
FC_UNITIRTXSET	Adh	4	0x01 =METHOD_START_IRTX	MSB usIRCode	LSB usIRCode	bChannel			IR transmitter unit	Response : EV_UnitDirectSet	Explicit invoke / SEGMENTED	L/N
		1	0x02 = METHOD_STOP_IRTX									
		4	0x03 = METHOD_PULS_IRTX	MSB usIRCode	LSB usIRCode	bChannel						
		2	0x04 =ATTR_CONFIG	bIRTXConfig								
		1	0x06 = FLUSH_DBASE									
FC_UNITREQUESTIRTXSTATUS	BFh	1	0x04 = CONFIG									
		1	0x05 = STATE									
FC_UnitRC5IRReceiverSet	B3h	3	0x01 = METHOD_INITIATE_CODE	bIRCodeIndex (0..63)	EV_UNITCONTROLTOGGLE EV_UNITCONTROLPULSTOGGLE EV_UNITCONTROLPULS	0/1 0/1 X			RCSRx unit	response with EV_DIRECTSET (if explicit invoke)	implicit/explicit	
		2	0x02 = ATTR_CONFIG	bRCSRxConfig 0 = RCS 1 = SONY								
FC_UnitRequestRC5IRReceiverStatus	B4h	1	0x02 = ATTR_CONFIG						RCSRx unit	response with EV_UNITRCSIRRECEIVERSTATUS	implicit/explicit	L/N
FC_UnitRequestAudioStatus (IR AUDIO all configurations or Bose audio)	A5h	1	0x01 = ATTR_SOURCE_ON						IR Audio unit in the Master LCD	response with EV_AUDIOSTATUS	implicit/explicit	L/N
		1	0x06 = ATTR_ONOFF									
		1	0x0D = ATTR_CONFIG									
		1	0x10 = ATTR_STATE									
		1	0x11 = ATTR_DISC (Bose)									
		1	0x12 = ATTR_TUNERFREQ (Bose)						Only with Bosers unit			
		1	0x13 = ATTR_VOLUME (Bose)									
FC_UnitRequestAudioStatus (MATRIX AUDIO HC4 and HC5)	A5h	2	0x01 ATTR_SOURCE_ON	bAudioDestNR					MATRIX Audio unit in the Master LCD	response with EV_AUDIOSTATUS	implicit/explicit	L/N
		2	0x06 ATTR_ONOFF	bAudioDestNR								
		2	0x0C ATTR_VOLUME	bAudioDestNR								
		1	0x0D ATTR_CONFIG									
		1	0x0E ATTR_STATE									
		1	0x16 ATTR_DEST_STATUS									
		1	0x17 ATTR_VOLUME STATUS									
FC_UnitAudioSet (IR AUDIO all configurations or Bose audio)	9Fh	1	0x00 = Use EV_UnitSelector data to set values						IR Audio unit in the Master LCD	response with EV_DIRECTSET (if explicit invoke)	implicit/explicit	L/N
		2	0x01 = ATTR_SOURCE_ON	bSourceSelection (0..5)								
		1	0x02 = METHOD_SOURCEINC									
		1	0x03 = METHOD_SOURCEDEC									
		1	0x04 = METHOD_CSON									
		1	0x05 = METHOD_OFF									
		2	0x06 = ATTR_ONOFF	0= OFF, 1 = ON								
		1	0x07 = METHOD_FSCSONOFF									
		1	0x08 = METHOD_VOLUP									
		1	0x09 = METHOD_VOLDOWN									
		1	0x0A = METHOD_FSVOLUPDOWN									
		1	0x0B = METHOD_PRESUP									
		1	0x0C = METHOD_PRESDOWN									
		2	0x0D = ATTR_CONFIG	bAudioConfig								
		2	0x0E = METHOD_SOURCE_FUNCTION	bFunctionNR (0..1)								
		2	0x0F = METHOD_DESTINATION_FUNCTION	bFunctionNR (0..1)								
		2	0x13 = ATTR_VOLUME (Bose)	bCurrentVolume								
		2	0x17 = METHOD_EXTRAIR	bFunctionNR(0 ... x)								
		3	0x18=METHOD_REMOTE_EXTRAIR	bKeyCode	bEvent							
		2	0x1F=ATTR_ZONE_SOURCE_ON	bSource								
		1	0x20=METHOD_PLAY									
		1	0x21=METHOD_STOP									
		1	0x22=METHOD_PAUSE									
		1	0x23=METHOD_FF									
		1	0x24=METHOD_RR									
		1	0x25=METHOD_FRWD									
		1	0x26=METHOD_FRWD									
										only for Bose V35. Bose: Only used to send local messages Bose: Only used to send local messages only for Bose V35. only for Bose V35. only for Bose V35. only for Bose V35. only for Bose V35. only for Bose V35. only for Bose V35.		
FC_UnitAudioSet (MATRIX AUDIO HC4 and HC5)	9Fh	1	0x00 = Use EV_UnitSelector data to set values						Matrix Audio unit in the Master LCD	response with EV_DIRECTSET (if explicit invoke)	implicit/explicit	L/N
		3	0x01 ATTR_SOURCE_ON	bAudioDestNR	bAudioSourceNR							
		2	0x02 METHOD_SOURCEINC	bAudioDestNR								
		2	0x03 METHOD_SOURCEDEC	bAudioDestNR								
		2	0x04 METHOD_CSON	bAudioDestNR								
		2	0x05 METHOD_OFF	bAudioDestNR								
		3	0x06 ATTR_ONOFF	bAudioDestNR	bState 0/1							
		2	0x07 METHOD_FSCSONOFF	bAudioDestNR								
		2	0x08 METHOD_VOLUP	bAudioDestNR								
		2	0x09 METHOD_VOLDOWN	bAudioDestNR								
		2	0x0A METHOD_VOLSTOP	bAudioDestNR								
		2	0x0B METHOD_FSVOLUPDOWN	bAudioDestNR								
		3	0x0C ATTR_VOLUME	bAudioDestNR	bDestinationVolume							
		2	0x0D ATTR_CONFIG	bAudioConfig								
		2	0x0E ATTR_STATE	bAudioStatus1								
		2	0x0F METHOD_PLAY	bAudioDestNR								
		2	0x10 METHOD_STOP	bAudioDestNR								
		2	0x11 METHOD_PAUSE	bAudioDestNR								
		2	0x12 METHOD_FF	bAudioDestNR								
		2	0x13 METHOD_RR	bAudioDestNR								
		2	0x14 METHOD_FRWD	bAudioDestNR								
		2	0x15 METHOD_FRWD	bAudioDestNR								
		6	0x16 ATTR_DEST_STATUS 22 / R data 5 bytes:	bCurrentDestination	bDestinationState0 bit 7 assigned/not assigned bit 6 on/off bit 5..0 source 0..63	bDestinationState1 bit 7 assigned/not assigned bit 6 on/off bit 5..0 source 0..63	bDestinationState2 bit 7 assigned/not assigned bit 6 on/off bit 5..0 source 0..63	bDestinationState3 bit 7 assigned/not assigned bit 6 on/off bit 5..0 source 0..63				
		5	0x17 ATTR_VOLUME STATUS	bDestinationVolume0								
		3	0x18 AUDIO_OBJECT_METHOD_EXTRAIR	bAudioDestNR	bDestinationVolume0 bIRCodeNr 0x00..0x18							

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TO DO

Message Code Macro name	Msg Code	length resultdata : DATA[0]	messagettransfer DATA[0] resultdata DATA[1]	messagettransfer DATA[1] resultdata DATA[2]	messagettransfer DATA[2] resultdata DATA[3]	messagettransfer DATA[3] resultdata DATA[4]	messagettransfer DATA[4] resultdata DATA[5] (description)	messagettransfer DATA[5] resultdata DATA[6] (description)	node/unit type	Remark	APDU service	Msg
FC_UnitRequestAVMatrixStatus	CBh	2	0x01 OBJECT ATTR CONNECTION 0x04 OBJECT ATTR VOLUME 0x05 OBJECT ATTR BASS 0x06 OBJECT ATTR TREBLE 0x0D AVMATRIX_OBJECT_ATTR_STATE 0x0E AVMATRIX_OBJECT_ATTR_CONFIG 0x0F OBJECT_CONFIG_NR_OF_DEST_ON_UNIT 0x10 OBJECT ATTR CONFIG NO OF UNITS	bMatrixOutput bMatrixOutput bMatrixOutput bMatrixOutput						Response : EV_UnitAVMatrixStatus	Explicit Invoke	L/N
FC_UnitAlarmSet	CCh	1 2	0x00 OBJECT USE SELECTOR DATA 0x01=OBJECT_METHOD_KEYPAD	0x00..0x09 = digits 0..9 0x0A = Full set / Arm 0x0B = Part set / Disarm 0x0C = C 0x0D = M bGroupNR (0x00..0x0F)								
		3	0x02=OBJECT_METHOD_ARM		0=Full set / Mode 0 1=Part set / Mode 1 2=Mode 2 3=Mode 3	char	char	char		Integra alarm uses arm modes 0-3		
		6 2	0x08=OBJECT_ATTR_REMOTECODE 0x09=OBJECT_ATTR_CONFIG	blindex (0...7) 0=Galaxy60 1=Galaxy500 2=Aritech						Since v1,7		
		1	0x0A=OBJECT_METHOD_LOGON									
FC_UnitRequestAlarmStatus	CDh	1 1 1 1 1	0x04=OBJECT_ATTR_GROUPSTATUS 0x05=OBJECT_ATTR_ALARMZONE 0x08=OBJECT_ATTR_REMOTECODE 0x09=OBJECT_ATTR_CONFIG 0x0B=OBJECT_ATTR_STATE	blindex (0...7)								
FC_UnitRequestSpeechcommStatus	C6h	1 1	0x0A=SPEECHCOMM_OBJECT_ATTR_CONFIG 0x0B=SPEECHCOMM_OBJECT_ATTR_STATE									
2. NODE OBJECT FUNCTIONS TO BIND OR DIRECT CALL												
FC_NODEMASTERLCDTAPISET	0xC7	1 or 2 1 1 1 1 1 1 1 or 3 or 4 1 3 1 1 1 1 1	0x01 = TAPI_METHOD_CALL 0x02 = TAPI_METHOD_HANGUP 0x03 = TAPI_METHOD_ACCEPT 0x01=TAPI_METHOD_CALL 0x02=TAPI_METHOD_HANGUP 0x03=TAPI_METHOD_ACCEPT 0x04=TAPI_METHOD_DOOR 0x05=TAPI_METHOD_DISCARD 0x06=TAPI_METHOD_REQUESTVIDEOSWITCH 0x07=TAPI_METHOD_CALLEND 0x08=TAPI_METHOD_CALLSTART 0x09=TAPI_METHOD_CALLBUSY 0x0A=TAPI_METHOD_CALLDISCARD	(outdoor device)						V66 nodes - Added outdoor device id.		
				(door id)		(outdoor device)	bState (1=connect,0=disconnect)			V66 Nodes - Added extra door + outdoor device.		
3. NODE/UNIT OBJECT CONFIGURATION FUNCTIONS												
FC_NodeRequestConfig	8Ah	0							node	Response : EV_NodeConfig	Explicit Invoke	L/N
FC_UnitRequestConfig	84h	0							unit	Response : EV_UnitConfig	Explicit Invoke	L/N
FC_UnitSetName	8Bh	2..N+2	object name index	char[n]	... char(n) ...				unit	Response : EV_UnitName	Implicit Invoke	L/N
FC_NodeSetName	8Ch	2..N+2	object name index	char[n]	... char(n) ...				node	Response : EV_NodeName	Implicit Invoke	L/N
FC_UnitRequestName	8Dh	1	object name index						unit	Response : EV_UnitName	Explicit Invoke	L/N
FC_NodeRequestName	8Eh	1	object name index						node	Response : EV_NodeName	Explicit Invoke	L/N
FC_NodeRequestStatus	8Fh	0							node		Explicit Invoke	L/N
FC_BrowseForNodes	90h	1	bNodeTypeMask (bit enable/disable) (see nodetypes)						node broadcast	Response Ev_NodeConfig	Implicit Invoke	N
FC_BrowseForNewNodes	91h	1	bNodeTypeMask (bit enable/disable) (see nodetypes)									
FC_NodeGetResources	9Dh	1	ResourceNR (0..2)						node	Response EV_Resources1/2	Explicit Invoke	L/N
FC_UnitConfig	93h	1	bUnitFlags bit 0 : NIC_NETWORK_CHANNEL_RETRY_ENABLED (DLL retry after timeout disabled or enabled) (factory settings enabled) bit 1 : NIC_NETWORK_CHANNEL_MT_RETRY (explicit_invoke retry disabled or enabled) (factory settings disabled) bit 2 : UNIT_SEND_AT_RESET (send status/events after reset) (factory settings disabled) bit4 : UNIT_EXCLUDE_FROM_BROADCAST (unitnr 0x0F)						unit	Response EV_UnitConfig	Explicit Invoke	L/N
FC_NodeConfig	94h	2	bNodeFlags bit 0 : NIC_NETWORK_CHANNEL_RETRY_ENABLED (DLL retry after timeout disabled or enabled) (factory settings enabled) bit 1 : NIC_NETWORK_CHANNEL_MT_RETRY (explicit_invoke retry disabled or enabled) (factory settings disabled) bit 3 : Master = 1, Slave = 0 (eg. for LCD node) bit 5.4 : LCD type bit 6 : matrixaudio=1; ifaudio=0	bDLAccessControl bit 7 : Auto_Retry 0 = Broadcast Unacknowledge DLL service 1 = Broadcast Addressed Unacknowledge DLL service (factory setting) bit 6 : 1 : Enable Immediate retry after 6-st UST symbol for ACK packets (no broadcast) 0 = disabled (factory setting) 1 = enabled bit 5.4 : Ch_Access_Period 00 = 0ms (factory setting) 01 = 100ms 10 = 200ms 11 = 300ms bit 3.2 : Access Control Bits, reserved, must be 00 bit 1.0 : Ch_Access_Num					unit	Response EV_NodeConfig	Explicit Invoke	L/N

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TO DO

Message Code Macro name	Msg Code	length resultdata : DATA[0]	message transfer DATA[0] resultdata DATA[1]	message transfer DATA[1] resultdata DATA[2]	message transfer DATA[2] resultdata DATA[3]	message transfer DATA[3] resultdata DATA[4]	message transfer DATA[4] resultdata DATA[5] (description)	message transfer DATA[5] resultdata DATA[6] (description)	node/unit type	Remark	APDU service	Msg
		1 PackLn 2 1 1	2 = Read Data 3 = Write Data 4 = Clear entry 5 = Done Write/Read 6 = Cancel Write/Read	Data Binding entry NR	Data	Data	Data	Data				
6. NODE/UNIT OBJECT PROTOCOL SPECIFIC FUNCTION												
FC_UnitReset FC_NodeReset	9Ah 9Bh	0 1	0x00 NORMAL_OPERATION 0x01 NODE_DEBUG_MODE 0x02 FACTORY_RESET & NORMAL_OPERATION (cleareeprom) 0x03 FACTORY_SETTINGS & NORMAL_OPERATION						unit node and units	response EV_Init response EV_Init	Implicit Invoke Implicit Invoke	L/N L/N
7. NODE OBJECT EEPROM SPECIFIC MANAGEMENT FUNCTIONS												
FC_NodeEraseEEPROM FC_DatabaseManagement	A0h B5h	0 2	0x00 OBJECT_ERASE 0x01 OBJECT_LOAD 0x02 OBJECT_SAVE 0x03 OBJECT_BACKUP	Database ID (0 ... 1) Database ID (0 ... 1) Database ID (0 ... 1) Database ID (0 ... 1)					node	Only in bNodeState == NODESTATUS_RUNNING response with EV_DatabaseManagement since v65.24	Implicit Invoke Explicit Invoke	N N
FC_NodeEEPROMRead FC_NodeEEPROMWrite	FEh FFh	2 3	Address MSB Address MSB	Address LSB Address LSB	Val							
FC_NodeRequestDLLStatus	95h	1	NIC error codes block NR							response with EV_NodeDLLStatus	Implicit Invoke	N
8. LCD NODE SPECIFIC PROTOCOL FUNCTIONS												
FC_NodeLCDMasterSet	A9h	2 5	0x00 = Weekschedule 0x01 = Alarm Set	bWeekScheduleNR (0..3) CODE MSB	CODE	CODE	CODE LSB	bAlarmStatus ALARM_ZONE_A = 0x01 ALARM_ZONE_B = 0x02 ALARM_ACTION = 0x10		response with EV_NodeLCDMasterStatus	Implicit Invoke	N
		6 6 6 6	0x07 = Registermap address 0x0000 - 0x00ff 0x08 = Registermap address 0x0100 - 0x01ff 0x09 = Registermap address 0x0200 - 0x02ff 0x07 = Registermap address 0x0300 - 0x03ff	RegisterAddress RegisterAddress RegisterAddress RegisterAddress	HMSB value HMSB value HMSB value HMSB value	HLSB value HLSB value HLSB value HLSB value	LMSB value LMSB value LMSB value LMSB value	LLSB value LLSB value LLSB value LLSB value				
FC_NodeLCDMasterRequestStatus	Abh	1 1 1 1 2 2 2 2	0x00 = Weekschedule 0x02 = Alarm Code 0x03 = Alarm Status 0x04 = Alarm Event 0x07 = Registermap address 0x0000 - 0x00ff 0x08 = Registermap address 0x0100 - 0x01ff 0x09 = Registermap address 0x0200 - 0x02ff 0x07 = Registermap address 0x0300 - 0x03ff	RegisterAddress RegisterAddress RegisterAddress RegisterAddress						response with EV_NodeLCDMasterStatus	Implicit Invoke	N
FC_NodeLCDMasterTimeDate	Aah	4 6	0x00 = LCDMASTER_TIMEDATE0 0x01 = LCDMASTER_TIMEDATE1	Hour DayNR	Minute WeekNR	Date QuarterNR	Month	Year			Implicit Invoke	N
FC_SlaveLCDInitiateBrowsing	9Eh	1	bNodeTypeMask (bit enable/disable) (see node types)									
FC_NodeSetMasterLCDNodeLA	96h	1	bNodeAddress (0..7)									
FC_NodeMasterLCDShutdown FC_NodeLCDShutdown	97h 9ch	0 0										
FC_NodeGatewaySet	F8h	1 2	0x02 = OBJECT_DATABASE_ERASE 0x03 = OBJECT_NOTIFY_DATABASE	0x00 = DATABASE_UPDATE_ERROR 0x01 = DATABASE_UPDATE_DONE 0x02 = DATABASE_UPDATE_STARTED					from PINInstall from Master LCD	response with EV_NODEDirectSet response with EV_NODEDirectSet or EV_MessageError	Explicit Invoke Explicit Invoke	
FC_NodeGatewayRequestStatus	F9h	1	0x01 = ATTR_STATE									
FC_LockLCD	FBh	1	LCD_TSLOCKED 0x02 LCD_LOCKED 0x01 LCD_UNLOCKED 0x00									
FC_UnLockLCD	Fah	0										
FC_GetMemoryData	FCh	6	memory address A31..24	memory address A23..16	memory address A15..8	memory address A7..0	w.Length 15..8	w.Length 7..0	LCD node	response with EV_MemoryDataDump	Implicit/explicit Invoke	N/L
FC_WriteMemoryData	FDh	Segmented	32 bit memory address followed by data						LCD node	response with EV_WriteMemoryDataDone	Implicit/explicit Invoke	N/L
FC_LCDPMDbaseAddNodeEntry	Bah	Segmented	PM database Node object (STRUCT_NODE) DATA = BIG ENDIAN						LCD node	response with EV_LCDPMDbaseAddNodeEntry	Explicit invoke	N/L
FC_LCDPMDbaseAddUnitEntry	BBh	Segmented	PM database Unit object (STRUCT_UNIT) DATA = BIG ENDIAN						LCD node	response with EV_LCDPMDbaseAddUnitEntry	Explicit invoke	N/L
FC_CHECKRC5CODE	C0h	3	MSB of usIRCode	LSB of usIRCode	bEvent 0 = RC5_TOGGLE_OFF_EVENT 1 = RC5_TOGGLE_ON_EVENT 2 = RC5_PULS_EVENT 3 = RC5_NO_EVENT				From RC5RX unit in std. Node to LCD node.	response with EV_RC5CODEMATCH if a match exist, otherwise nothing !!	Implicit invoke	N/L
FC_GETIRCODE	Beh	2	MSB of usRequestCode	LSB of usRequestCode					From RC5RX unit in std. Node to LCD node. LCD node.	response with EV_IRCODEDATA or EV_IRCODEDATABASEERROR	APDU EXPLICIT RETRY	N/L
FC_LCDINFODIALOG	BCh	Segmented	textstring							Explicit invoke		N/L
FC_LCDBUILDSCREENMAP	C8h	4 5	SCREENMAP_INIT SCREENMAP_INIT	bScreenmapID bScreenmapID	MSB usNROfEntries MSB usNROfEntries	LSB usNROfEntries LSB usNROfEntries	bStartPageNR			Vanaf binding software Version 0x0571		

TO DO

Message Code	Msg Code	bLength resultdata : DATA[0]	messagetransfer DATA[0] resultdata DATA[1]	messagetransfer DATA[1] resultdata DATA[2]	messagetransfer DATA[2] resultdata DATA[3]	messagetransfer DATA[3] resultdata DATA[4]	messagetransfer DATA[4] resultdata DATA[5] (description)	messagetransfer DATA[5] resultdata DATA[6] (description)	node/unit type	Remark	APDU service	Msq
	6		SCREENMAP_ENTRY	bScreenmapID	bPageNR	bButtonNR	bNodeLogicalAddress	bUnitAddress				
9. Reserved Functions Modem v65.24												
FC_NODEMANAGEAVIRDEVICES	80h										Not reserved in modem	
FC_RESERVED81	81h										Explicit invoke	
FC_RESERVED82	82h										Explicit invoke	
FC_RESERVED92	92h										Explicit invoke	
FC_UNITREQUESTLCDSTATUS	A4h										Explicit invoke	
FC_UNITFEEDBACKCONFIG	Ach										Explicit invoke address	
FC_NODEAVIRCONNECTIONMATRIXSET	B0h										Explicit invoke	
FC_NODEREQUESTAVIRMATRIXSTATUS	B1h										Explicit invoke	
FC_NODEAVIRDEVICESENDIRCODE	B2h										Explicit invoke	
FC_TEMP1	Ceh										Explicit invoke	
FC_TEMP2	CFh										Explicit invoke	
FC_TEMP3	D0h										Explicit invoke	
FC_TEMP4	D1h										Explicit invoke	
Remark :												
N : Depends on the MaxMessageBufferLength got from EV_NodeResources												
The Length of the message data part should not exceed MaxMessageBufferLength												
Message domain column N : Network, L : Local												
MACROS USED WITH EVENTS/FUNCTIONS												
NodeTypes												
NODE_PIN_STD_UNITS	0x01	// Node with pure Dim, Control, Sens, Switch, ... units...										
NODE_PIN_ROUTER_TIN	0x02	// Route between PL and TP.										
NODE_PIN_GATEWAY	0x04											
NODE_PIN_MODEM	0x08											
NODE_RESERVED	0x10	(Herbruiken)										
NODE_LCD	0x20											
Unit Types												
/* unit types*/												
#define UNIT_TYPE_DIM	0x01											
#define UNIT_TYPE_SWITCH	0x02											
#define UNIT_TYPE_CONTROL	0x03											
#define UNIT_TYPE_SENS	0x04											
#define UNIT_TYPE_AUDIO	0x05											
#define UNIT_TYPE_LCD	0x06	(Herbruiken)										
#define UNIT_TYPE_LCD_VIRTUAL	0x07											
#define UNIT_TYPE_DUOSWITCH	0x08											
#define UNIT_TYPE_SELECTOR	0x09											
#define UNIT_TYPE_BOSERS	0x0A											
#define UNIT_TYPE_AVMATRIX	0x0B											
#define UNIT_TYPE_IRTX	0x0C											
#define UNIT_TYPE_RCSR	0x0D											
#define UNIT_TYPE_VIDEOMUX	0x0E											
#define UNIT_TYPE_PS2INTELLIMOUSE	0x0F											
#define UNIT_TYPE_SPEECHCOMM	0x10	(Herbruiken)										
#define UNIT_TYPE_SPMATRIX	0x11											
#define UNIT_TYPE_ALARM	0x12											
Remark:												
When using CAN, broadcast messages node to unit ! are limited to a data of 3 bytes												
Segmented messages can have a length of max 255*7+6 = 1791 bytes												